

Module 4, Assignment 4: Output Check

Ellen Bledsoe, Lily McMullen

Problem Set 1

Flipper length range and mean by species (Dream island):

```
# A tibble: 2 x 4
  species    min_flipper max_flipper mean_flipper
  <fct>         <int>         <int>         <dbl>
1 Adelie         178          208          190.
2 Chinstrap      178          212          196.
```

T-test:

Welch Two Sample t-test

```
data: flipper_length_mm by species
t = -4.937, df = 120.37, p-value = 2.581e-06
alternative hypothesis: true difference in means between group Adelie and group Chinstrap is
95 percent confidence interval:
 -8.534213 -3.648561
sample estimates:
 mean in group Adelie mean in group Chinstrap
          189.7321          195.8235
```

Problem Set 2

Mean and standard deviation of bill length and flipper length:

```
# A tibble: 1 x 4
  mean_bill sd_bill mean_flipper sd_flipper
    <dbl>   <dbl>      <dbl>      <dbl>
1    43.9    5.46        201.        14.1
```

Correlation coefficient and r-squared:

```
[1] 0.6561813
```

```
[1] 43.0574
```

Regression model:

Call:

```
lm(formula = bill_length_mm ~ flipper_length_mm, data = penguins_noNA)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-8.5792 -2.6715 -0.5721  2.0148 19.1518
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   -7.26487    3.20016   -2.27   0.0238 *
flipper_length_mm 0.25477    0.01589   16.03  <2e-16 ***
---

```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 4.126 on 340 degrees of freedom

Multiple R-squared: 0.4306, Adjusted R-squared: 0.4289

F-statistic: 257.1 on 1 and 340 DF, p-value: < 2.2e-16

Multiple regression with species interaction:

Call:

```
lm(formula = bill_length_mm ~ flipper_length_mm * species, data = penguins_noNA)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
```

-6.6977 -1.7046 0.0596 1.5571 12.4394

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	13.58714	6.05061	2.246	0.025380	*
flipper_length_mm	0.13269	0.03183	4.168	3.91e-05	***
speciesChinstrap	-7.99376	10.48117	-0.763	0.446190	
speciesGentoo	-34.32335	9.81983	-3.495	0.000537	***
flipper_length_mm:speciesChinstrap	0.08813	0.05405	1.631	0.103915	
flipper_length_mm:speciesGentoo	0.18152	0.04775	3.801	0.000171	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.55 on 336 degrees of freedom

Multiple R-squared: 0.7851, Adjusted R-squared: 0.7819

F-statistic: 245.5 on 5 and 336 DF, p-value: < 2.2e-16

Problem Set 3

Body mass summary by species:

```
# A tibble: 3 x 4
  species min_mass max_mass mean_mass
  <fct>    <int>    <int>    <dbl>
1 Adelie   2850     4775     3701.
2 Chinstrap 2700     4800     3733.
3 Gentoo   3950     6300     5076.
```

Converted to kilograms:

```
[1] 2.85 2.70 3.95
[1] 4.775 4.800 6.300
[1] 3.700662 3.733088 5.076016
```

```
# A tibble: 3 x 4
  species min_mass max_mass mean_mass
  <fct>    <dbl>    <dbl>    <dbl>
1 Adelie   2.85     4.78     3.70
2 Chinstrap 2.7      4.8      3.73
3 Gentoo   3.95     6.3      5.08
```

ANOVA:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
species	2	146864214	73432107	343.6	<2e-16 ***
Residuals	339	72443483	213698		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
2 observations deleted due to missingness

Pairwise comparison:

Tukey multiple comparisons of means
95% family-wise confidence level

Fit: aov(formula = body_mass_g ~ species, data = penguins)

\$species		diff	lwr	upr	p adj
Chinstrap-Adelie		32.42598	-126.5002	191.3522	0.8806666
Gentoo-Adelie		1375.35401	1243.1786	1507.5294	0.0000000
Gentoo-Chinstrap		1342.92802	1178.4810	1507.3750	0.0000000